

AL-AZHARI LIBRARY — FINAL IMPLEMENTATION PLAN COMPLIANCE AUDIT

Repository-Level Verification Report

Auditor Role: Senior Backend Architect + QA Auditor + Repository Compliance Reviewer
Authoritative Plan: `documentation/AL-AZHARI-LIBRARY-BACKEND-IMPLEMENTATION-PLAN.pdf` (64 pages)
Database Plan: `documentation/AL-AZHARI-LIBRARY-MongoDB-Implementation-Plan.pdf`
Target Repository: AL-Azhary web store / Back-end
Execution Baseline: Phases 0–18 Complete
Audit Rule: Strict evidence-based evaluation (Requirement → Implementation → Route/Service/Schema → Test/Evidence).
No assumptions based on phase completion reports alone.

1. Executive Summary

A comprehensive, line-by-line compliance audit was conducted comparing the 64-page **Backend Implementation Plan** against the actual codebase in `Back-end/`.

Total plan requirements audited: 68
PASS: 55
FAIL: 4
NEEDS VERIFICATION: 3
OPEN DECISION: 6 (Preserved system-wide across 24 specific open decision IDs)

Key High-Level Findings

- 1. **Core E-Commerce, Services, Auth & Security Foundations: 100% COMPLIANT.** Authentication, addresses, product catalog, categories, cart versioning/merge, inventory reservation/ledger, order state machines, manual payment proofs via Cloudinary, shipping rules, coupons, student services, custom quotations, returns/refunds, in-app notifications, outbox worker, audit logging, analytics reports, layered rate limiting, NoSQL injection prevention, and deployment infrastructure are thoroughly implemented, tested (137 test suites, 938 tests, 100% pass), and robust.
- 2. **Area A — OpenAPI + Postman: FAIL.**
 - `src/config/openapi.ts` defines only **11 endpoints** out of the **107 routes** registered in Express.
 - Swagger UI is not served on any endpoint.
 - Postman collection is **missing from the repository** entirely.
 - The required CI automated synchronization check (failing if an Express route is missing from OpenAPI) does not exist.
- 3. **Area B — Pre-orders: FAIL.**
 - While `PreorderModel` exists in `src/modules/preorders/models/preorder.model.ts` (with an index created in Phase 15 to feed the `preorder-demand` analytics report), the core pre-order business endpoints (`POST /api/v1/products/:slug/pre-orders` , `GET /api/v1/pre-orders` , and `POST /api/v1/admin/pre-orders/:reference/accept` as mandated by Plan Section 48.7) are **NOT implemented**.
 - The files `preorder.controller.ts` , `preorder.repository.ts` , `preorder.routes.ts` , `preorder.schema.ts` , and `preorder.service.ts` are **0-byte empty stub files**.
 - Pre-order routes are not mounted in `src/app.ts` .
- 4. **Area C — CMS / Content: PASS.**
 - Fully implemented in `src/modules/content/` (`ContentModuleModel` , `ContentService` , `ContentController`).
 - Public route `GET /api/v1/content/home` with date filtering, and admin CRUD `/api/v1/admin/content/*` with `content.write` permission.
 - Verified that the seasonal/content layer cannot mutate taxonomy (verified in `tests/api/content-api.test.ts`).

5. Area D — WhatsApp Integration: FAIL.

- `src/integrations/whatsapp` contains only a `.gitkeep` file.
- The mandated utility function `buildWhatsAppUrl(phone, context)` (Plan Section 38.1) is **not implemented**.
- No tests exist verifying WhatsApp URL encoding and secret redaction (`WA-001-003`).

6. Area E — Performance / Query Requirements: PASS.

- Bounded pagination enforced globally (`parsePagination` : default 20, max 100).
- Zero occurrences of `populate()` in hot transactional paths.
- External network calls (Cloudinary, SMTP, Socket.IO) are strictly executed asynchronously via the Outbox worker and never held inside MongoDB multi-document transactions.

2. FINAL VERDICT

VERDICT: NOT FULLY COMPLIANT

Justification

While the core commerce, service, and security architecture is production-grade, the repository fails four explicit plan requirements:

1. **OpenAPI Specification incompleteness** (documents only ~10% of routes; no CI sync check).
2. **Missing Postman collection artifact**.
3. **Unimplemented Pre-orders feature module** (0-byte stubs, unmounted routes, missing endpoints).
4. **Unimplemented WhatsApp URL helper / integration adapter** (`src/integrations/whatsapp` empty).

3. MAIN COMPLIANCE MATRIX

Plan Requirement Group	Requirement ID	File / Route	Implementation Evidence	Test / Evidence
Architecture & Runtime	ARCH-001-006	server.ts, app.ts	Modular monolith, Node.js 20+, Express, TypeScript, Mongoose, REST /api/v1	app-bootstrap.test.t health-smoke.test.ts
Authentication & Sessions	AUTH-001-008	auth.routes.ts	Register, login, refresh rotation, logout, email verify, forgot/reset password, Argon2id, hashed tokens	auth-login.test. auth-refresh.test.ts , auth-logout.test
User & Address Management	USER-001-005	address.routes.ts, me.routes.ts	Profile CRUD, address book, default address management, ownership enforcement	address-crud.test.ts , address-ownership.test.t me.test.ts
Catalog & Categories	CAT-001-008	product.routes.ts, category.routes.ts	Books-first categories, product variants, normalized search, admin catalog CRUD	product-api.test category-api.test.ts , sea api.test.ts
Shopping Cart	CART-001-005	cart.routes.ts	Guest/customer cart persistence, optimistic versioning, cart merge	cart-user.test.t cart-guest.test. cart-merge.test.
Inventory & Concurrency	INV-001-008	inventory.service.ts	Atomic conditional reservation, ledger audit trail, final deduction on fulfillment	inventory-concurrency.test inventory-reservation.test
Checkout & Shipping Rules	CHK-001-012 , SHIP-001-009	checkout.routes.ts, shipping.service.ts	Shipping estimate by governorate/city/area, pickup vs delivery, order transaction	order-checkout.test.ts shipping-service.test.ts
Order Management	ORD-001-009	order.routes.ts, admin-order.routes.ts	Order lifecycle state machine, guest access token, customer cancellation, COD confirmation	order-api.test.t order-admin-api.test.ts , ord lifecycle.test.t
Manual Payments & Proofs	PAY-001-008	payment.routes.ts, admin-payment.routes.ts	Private Cloudinary upload config, proof submission, admin confirm/reject/request-new-proof, signed URLs	payment-api.test payment-admin-api.test.ts , med security.test.ts
Coupons	COUP-001-005	coupon.routes.ts	Fixed/percentage discounts, min order amount, redemption tracking, single-use per customer	coupon-api.test. coupon-service.test.ts

Plan Requirement Group	Requirement ID	File / Route	Implementation Evidence	Test / Evidence
Student Services	SRV-001-008	service.routes.ts	Configurable dynamic form fields, no-attachment enforcement, service request lifecycle	service-api.test service-schema.test.ts
Custom Quotations	QUOTE-001-005	quotation.routes.ts	Admin quote issuance, customer accept/reject, payment creation upon quote acceptance	quotation-api.test.ts , service-quotations-concurrency.test
Returns & Refunds	RET-001-004 , REF-001-003	returns.routes.ts	Return request eligibility, admin approve/reject, manual refund completion	return-api.test. return-concurrency.test
Pre-orders	PRE-001-007	src/modules/preorders/*	Plan Section 31 & 48.7 endpoints (POST /products/:slug/pre-orders , GET /pre-orders , POST /admin/pre-orders/:reference/accept)	Stubs only; no routes or controllers
CMS / Content	CMS-001-004	content.routes.ts	Content modules, seasonal display windows, bilingual titles/bodies, taxonomy isolation	content-api.test
WhatsApp Integration	WA-001-003	src/integrations/whatsapp	Plan Section 38 buildWhatsAppUrl(phone, context) helper, encoding, no-secret verification	Directory contains only .gitkeep
Notifications & Realtime	NOT-001-004 , RT-001-003	notification.routes.ts, socket/	Persisted MongoDB notifications, unread count, Socket.IO rooms with JWT auth	notification-api.test.ts , realtime-socket.test.ts , socket-security.test.ts
Outbox & Jobs	OUTBOX-001-005 , JOB-001-004	workerts, cron.ts	Transactional outbox, distributed lease recovery, exponential backoff, node-cron cleanup	outbox-worker-lifecycle.test.t outbox-claiming-concurrency.test
Reports & Analytics	REP-001-002	report.routes.ts	8 aggregation reports (orders, revenue, outside-gena, payment methods, preorder demand, etc.)	report-admin-api.test.ts
Audit Logging	AUD-001-004	audit.routes.ts	Append-only immutable audit logs, sensitive data redaction, admin query interface	audit-admin-api.test.ts , audit-immutability.test audit-redaction.test.t

Plan Requirement Group	Requirement ID	File / Route	Implementation Evidence	Test / Evidence
Security Hardening	SEC-001-006	Global Middleware & Config	Helmet, strict CORS, NoSQL sanitizer, 6 rate limit buckets, request ID, safe errors	14 security test suites (55 tests)
OpenAPI Documentation	Plan Sec. 48 & 49	openapi.ts	Only 11 of 107 routes documented; Swagger UI unmounted; no CI route-sync test	No contract/sync in tests/
Postman Collection	Plan Sec. 49 & 66	Repository Root & postman/	Postman collection file exported from API contract	File does not exist anywhere in repo
Database Migrations	Plan Sec. 10 & 58	migrations/	13 versioned migrations, idempotent runner CLI, zero startup destructive DDL	database-migrations-smoke.test.ts , migrations-regression.test.ts
Deployment Readiness	Phase 18 / Sec. 56-57	docs/PHASE-18-DEPLOYMENT.md	Single process runtime, Hostinger guide, Atlas TLS, health probes, graceful shutdown	tests/deployment suites, 25 tests)

4. AREA A — OPENAPI + POSTMAN MATRIX

Comparison between actual Express application routes, OpenAPI 3.0 specification (`src/config/openapi.ts`), and Postman collection:

Route Path	Method	Actual Express Route	Ir openap
/api/v1/auth/register	POST	Yes (auth.routes.ts)	NO
/api/v1/auth/login	POST	Yes (auth.routes.ts)	NO
/api/v1/auth/refresh	POST	Yes (auth.routes.ts)	NO
/api/v1/auth/logout	POST	Yes (auth.routes.ts)	NO
/api/v1/auth/verify-email	POST	Yes (auth.routes.ts)	NO
/api/v1/auth/forgot-password	POST	Yes (auth.routes.ts)	NO
/api/v1/auth/reset-password	POST	Yes (auth.routes.ts)	NO
/api/v1/me	GET, PATCH	Yes (me.routes.ts)	NO
/api/v1/categories	GET	Yes (category.routes.ts)	NO
/api/v1/products	GET	Yes (product.routes.ts)	NO
/api/v1/products/:slug	GET	Yes (product.routes.ts)	NO
/api/v1/search	GET	Yes (search.routes.ts)	NO
/api/v1/content/home	GET	Yes (content.routes.ts)	NO
/api/v1/cart	GET	Yes (cart.routes.ts)	NO
/api/v1/cart/items	POST	Yes (cart.routes.ts)	NO
/api/v1/cart/items/:itemId	PATCH, DELETE	Yes (cart.routes.ts)	NO
/api/v1/checkout/shipping-estimate	POST	Yes (checkout.routes.ts)	NO
/api/v1/orders	POST	Yes (order.routes.ts)	NO
/api/v1/orders/:reference	GET, PATCH	Yes (order.routes.ts)	NO
/api/v1/orders/:reference/cancel	POST	Yes (order.routes.ts)	NO
/api/v1/orders/:reference/confirm-cod	POST	Yes (order.routes.ts)	NO
/api/v1/orders/:reference/payment	GET	Yes (payment.routes.ts)	YES
/api/v1/orders/:reference/payment-proof/upload-config	POST	Yes (payment.routes.ts)	YES
/api/v1/orders/:reference/payment-proofs	POST	Yes (payment.routes.ts)	YES

Route Path	Method	Actual Express Route	Ir openap
/api/v1/admin/payments	GET	Yes (admin-payment.routes.ts)	YES
/api/v1/admin/payments/:paymentId	GET	Yes (admin-payment.routes.ts)	YES
/api/v1/admin/payments/:paymentId/confirm	POST	Yes (admin-payment.routes.ts)	YES
/api/v1/admin/payments/:paymentId/reject	POST	Yes (admin-payment.routes.ts)	YES
/api/v1/admin/payments/:paymentId/request-new-proof	POST	Yes (admin-payment.routes.ts)	YES
/api/v1/admin/payments/:paymentId/proofs/:submissionNumber/signed-url	GET	Yes (admin-payment.routes.ts)	YES
/api/v1/admin/audit-logs	GET	Yes (audit.routes.ts)	YES
/api/v1/admin/reports/:report	GET	Yes (report.routes.ts)	YES
/api/v1/services	GET	Yes (service.routes.ts)	NO
/api/v1/services/:slug	GET	Yes (service.routes.ts)	NO
/api/v1/services/:slug/requests	POST	Yes (service.routes.ts)	NO
/api/v1/service-requests/:reference	GET	Yes (service.routes.ts)	NO
/api/v1/service-requests/:reference/quotation/accept	POST	Yes (quotation.routes.ts)	NO
/api/v1/service-requests/:reference/quotation/reject	POST	Yes (quotation.routes.ts)	NO
/api/v1/orders/:orderReference/returns	POST	Yes (returns.routes.ts)	NO
/api/v1/returns	GET	Yes (returns.routes.ts)	NO
/api/v1/admin/returns	GET	Yes (returns.routes.ts)	NO
/api/v1/admin/returns/:reference/approve	POST	Yes (returns.routes.ts)	NO
/api/v1/admin/returns/:reference/reject	POST	Yes (returns.routes.ts)	NO
/api/v1/admin/refunds/:id/complete	POST	Yes (returns.routes.ts)	NO

Route Path	Method	Actual Express Route	Ir openap
(Remaining 63 Admin/Public Routes)	Various	Yes (107 routes total)	NO

5. AREA B — PRE-ORDER MATRIX

Detailed audit of the pre-order capabilities required by Plan Section 31 and Section 48.7:

Plan Requirement	Model	Service	Route	Test	
Demand Persistence	PreorderModel exists (preorder.model.ts)	None (preorder.service.ts is 0 bytes)	None	report-admin-api.test.ts (seeds dummy docs)	NEE VER (Mo use rep
Request Pre-order (POST /products/:slug/pre-orders)	Schema has productId , variantId , customerSnapshot , quantity	None	Missing from Express app	None	FAIL
Customer Pre-order Listing (GET /pre-orders)	Schema has customerId index	None	Missing from Express app	None	FAIL
Admin Accept Pre-order (POST /admin/pre-orders/:reference/accept)	None (reference field absent in PreorderModel)	None	Missing from Express app	None	FAIL
Pre-order Lifecycle Enum (requested , admin_review , accepted , rejected , payment_pending , confirmed , available , fulfilled , cancelled)	Discrepancy: model uses ['pending', 'confirmed', 'cancelled', 'fulfilled']	None	None	None	FAIL
Out-of-stock & Eligibility Validation	preOrderEligible boolean exists on Product & Variant	None	None	variant.test.ts checks product flag only	NEE VER

6. AREA C — CMS / CONTENT MATRIX

Detailed audit of CMS and seasonal content modules required by Plan Section 48.2 and Section 63:

Requirement	Model	Service	Route
Content Module Schema	ContentModuleModel (content-module.model.ts)	ContentService (content.service.ts)	src/modules/content/routes/content.
Public Homepage Content (GET /api/v1/content/home)	Date-bounded active query with startsAt / endsAt	getHomeContent()	GET /api/v1/content/home
Admin Content Management (GET, POST, PATCH, DELETE /api/v1/admin/content)	Validated key uniqueness, displayOrder, active flag	listAdminContent() , createContentModule() , updateContentModule() , deleteContentModule()	/api/v1/admin/content/*
Taxonomy Preservation (Seasonal layer cannot mutate product/category collections)	Ref-only (productId , categoryId)	getHomeContent() projects linked products/categories safely without mutating core collections	Read-only projection
Bilingual Localization	title.ar (required), title.en (optional), body.ar , body.en	Full projection to safe public / admin DTOs	Handled

7. AREA D — WHATSAPP MATRIX

Detailed audit of WhatsApp link generation and redaction required by Plan Section 38 and Section 63:

Plan Requirement	Implementation	Security Check	Test	Status
buildWhatsAppUrl(phone, context) Helper (Plan Sec. 38.1)	Missing: src/integrations/whatsapp has only .gitkeep	Not applicable (unimplemented)	None	FAIL
No Authoritative State Mutation	Chat links do not trigger database transactions or mutate order/service state	Verified: chat is treated purely as an external communication channel	None	PASS (Design-level)
URL Encoding & PII/Secret Redaction (WA-001-003)	None	None	None	FAIL
Contextual Deep Linking (product, orderReference, serviceReference)	None	None	None	FAIL

8. AREA E — PERFORMANCE / QUERY MATRIX

Audit of query performance, pagination, population, and database efficiency:

Performance Requirement	Location	Current Implementation	Evidence	Status
Bounded Pagination	pagination.ts	parsePagination : defaultLimit=20 , maxLimit=100 . Enforced on all list queries.	pagination.test.ts	PASS
No populate() in Hot Paths	Repository-wide	0 occurrences of .populate(found across all domain services. Data snapshots and IDs used instead.	Codebase scan	PASS
Lean Projections	Read repositories	Projections mapped through pure functions (toSafePublicProduct , toSafeAdminOrder , etc.) and .lean() in outbox/audit/migrations.	product.projection.ts , audit.repository.ts	PASS
No Network Calls in Transactions	Services & Transactions	withTransaction blocks exclusively execute MongoDB queries and Outbox insertions. Cloudinary, SMTP, and Socket.IO calls are deferred to outbox worker.	order.service.ts , payment.service.ts	PASS
Bounded Outbox Processing	outbox-worker.ts	Bounded batch size (OUTBOX_BATCH_SIZE=20), atomic claim via findOneAndUpdate , lease duration (OUTBOX_LEASE_MS=60000).	outbox-worker- lifecycle.test.ts , outbox-claiming- concurrency.test.ts	PASS
NoSQL Operator Sanitizer	nosql- injection.middleware.ts	Recursive scanner blocking all \$ operators and dot- notation paths across params, query, body.	nosql-injection.test.ts (7 tests)	PASS

9. DEFINITION OF DONE MATRIX

Evaluation against Plan Section 62 (Definition of Done):

Definition of Done Item	Evidence	Status
Domain boundary and source requirement IDs recorded	Recorded in <code>ARCHITECTURE.md</code> , module directory structures, and phase docs.	PASS
Request schemas, response DTOs, and error codes exist	Zod schemas and typed <code>AppError</code> error codes exist for all implemented modules (missing for preorders & whatsapp).	PARTIAL
Authentication, permission, ownership, and redaction tested	Tested in <code>tests/api/rbac-matrix.test.ts</code> , <code>tests/api/idor-regression.test.ts</code> , <code>tests/security/</code> .	PASS
State transitions are explicit and invalid transitions fail	State machines implemented with transition matrices for orders, payments, quotations, returns.	PASS
Transactions/idempotency applied where required	Multi-document transactions via <code>withTransaction</code> ; partial unique indexes for idempotency keys.	PASS
Audit, notification, and outbox effects implemented	Audit entries, in-app notifications, and outbox events written inside transaction boundaries.	PASS
Indexes and migration changes are versioned	13 versioned migrations registered and tracked in <code>__migrations</code> collection.	PASS
Open decisions are not silently hard-coded	Preserved in configuration, defaults, or neutral error responses.	PASS
Unit, integration, API, concurrency, and security tests pass	137 test suites, 938 tests running and passing (100% pass rate).	PASS
OpenAPI and Postman documentation match the route	OpenAPI has only 11 routes; Postman collection is missing.	FAIL
Logs do not expose secrets or unnecessary PII	Pino redaction configured; verified in <code>tests/security/error-redaction.test.ts</code> .	PASS
Health/readiness and deployment behavior verified	<code>/health/live</code> , <code>/health/ready</code> , and graceful shutdown verified in <code>tests/deployment/</code> .	PASS

10. ACCEPTANCE CRITERIA MATRIX

Evaluation against Plan Section 63 (Acceptance Criteria):

Acceptance Criterion	Repository Evidence	Test Evidence	Status
Guest Browse & Checkout (Add products/variants, pickup/delivery, shipping estimate, submit order)	<code>cart.routes.ts</code> , <code>checkout.routes.ts</code> , <code>order.service.ts</code>	<code>cart-guest.test.ts</code> , <code>order-checkout.test.ts</code>	PASS
Customer Browse, Order & History (Customer order history viewing)	<code>order.routes.ts</code> (<code>GET /api/v1/orders/:reference</code>)	<code>order-api.test.ts</code>	PASS
Pending Review Initial State (No stock deducted on order creation)	<code>order.service.ts</code> sets <code>status: 'pending_review'</code> ; stock untouched	<code>order-lifecycle.test.ts</code>	PASS
Atomic Inventory Reservation on Admin Acceptance	<code>order.service.ts</code> calls <code>inventoryService.reserveStock()</code> inside MongoDB transaction	<code>inventory-concurrency.test.ts</code> , <code>order-concurrency.test.ts</code>	PASS
COD Customer Confirmation Workflow	<code>order.service.ts</code> transitions <code>accepted COD order</code> to <code>confirmed</code> upon customer action	<code>order-lifecycle.test.ts</code>	PASS
Manual Payment Proof & Review (Screenshot proof required for non-COD; admin confirm/reject/request new proof)	<code>payment.service.ts</code> , <code>admin-payment.routes.ts</code>	<code>payment-proof-lifecycle.test.ts</code> , <code>payment-api.test.ts</code>	PASS
Private Payment Proofs (No direct public URLs; signed short-lived URLs for authorized admin only)	Private Cloudinary folder; <code>GET /admin/payments/:id/proofs/:num/signed-url</code>	<code>media-security.test.ts</code>	PASS
Fulfillment & Final Stock Deduction (Stock deducted exactly once at delivery/pickup)	<code>order.service.ts</code> calls <code>inventoryService.consumeStock()</code> at pickup/delivery	<code>inventory-lifecycle.test.ts</code>	PASS
Services Boundary (Discoverable, requestable, never cart items, config-driven fields, no attachments)	<code>service.service.ts</code> , <code>service.schema.ts</code> rejects file fields	<code>service-api.test.ts</code> , <code>service-schema.test.ts</code>	PASS
Quotations Workflow (Admin quote, customer accept/reject, payment gated until acceptance)	<code>quotation.service.ts</code> , <code>quotation.routes.ts</code>	<code>quotation-api.test.ts</code> , <code>service-quotation-concurrency.test.ts</code>	PASS
Role-Based Access Control (Admin bound to RBAC; Owner holds unrestricted access)	<code>rbac.middleware.ts</code> , <code>roles.service.ts</code>	<code>rbac-matrix.test.ts</code> , <code>rbac.test.ts</code>	PASS

Acceptance Criterion	Repository Evidence	Test Evidence	Status
Audit Logging (Audit records for critical mutations without exposing secrets)	<code>audit.service.ts</code> , <code>audit.middleware.ts</code>	<code>audit-admin-api.test.ts</code> , <code>audit-immutability.test.ts</code>	PASS
Delivery Resilience (Socket.IO / email failure does not roll back committed DB state)	Outbox pattern handles external delivery asynchronously	<code>outbox-worker-lifecycle.test.ts</code>	PASS
Token & Session Security (Argon2id, refresh token rotation, session revocation)	<code>jwt.service.ts</code> , <code>token.service.ts</code> , <code>auth.service.ts</code>	<code>auth-refresh.test.ts</code> , <code>auth-logout.test.ts</code> , <code>token-security.test.ts</code>	PASS
Safe Error Handling (Stable error code envelope without leaking internals)	<code>error.middleware.ts</code> , <code>AppError.ts</code>	<code>error-redaction.test.ts</code> , <code>errors.test.ts</code>	PASS
Production Environment Guardrails (Rejection of default secrets, insecure cookies, wildcard CORS)	<code>src/config/env.ts</code>	<code>env-security.test.ts</code> , <code>config-validation-smoke.test.ts</code>	PASS

11. FAILURES

Failure 1: Incomplete OpenAPI Specification

- **Requirement:** Plan Section 48 & 49 mandates comprehensive OpenAPI 3.0 documentation covering all `/api/v1` routes, request/response DTOs, parameters, security schemes, and error codes.
- **Expected:** All 107 registered Express endpoints documented in OpenAPI format with complete contract specifications.
- **Actual:** `src/config/openapi.ts` defines only 11 paths (all relating to payments, audit logs, and reports). 96 routes are entirely absent. Swagger UI is not mounted on any route.
- **File:** `Back-end/src/config/openapi.ts`
- **Why it fails:** OpenAPI configuration was incrementally authored during payment/audit phases and never updated to include the full API catalog.
- **Recommended fix:** Expand `src/config/openapi.ts` to document all remaining 96 endpoints (or generate dynamically from Zod schemas), and optionally mount `swagger-ui-express` protected behind admin authentication. Add an automated CI contract test comparing registered Express routes against OpenAPI paths.

Failure 2: Missing Postman Collection Artifact

- **Requirement:** Plan Section 49.1 and Section 66 (Technical Decision #28) explicitly mandates maintaining an exported Postman collection corresponding to the API endpoint contracts.
- **Expected:** A valid Postman Collection v2.1 JSON file located in `postman/` or project documentation directory with environment variables and sample requests.
- **Actual:** No Postman collection file exists in the repository.
- **File:** Repository root / `postman/`
- **Why it fails:** Postman collection was never created or committed.

- **Recommended fix:** Export an OpenAPI-derived Postman collection into `postman/al-azhari-library.postman_collection.json` with associated environment template.

Failure 3: Unimplemented Pre-orders Module

- **Requirement:** Plan Section 31 and Section 48.7 mandate pre-order functionality (PRE-001-007) allowing customers to request out-of-stock items, administrators to review and accept pre-orders, and status tracking across the pre-order lifecycle.
- **Expected:** Functional endpoints (`POST /api/v1/products/:slug/pre-orders` , `GET /api/v1/pre-orders` , `POST /api/v1/admin/pre-orders/:reference/accept`), service logic, and route mounting.
- **Actual:**
 - `preorder.controller.ts` , `preorder.repository.ts` , `preorder.routes.ts` , `preorder.schema.ts` , and `preorder.service.ts` are 0-byte empty stub files in `src/modules/preorders/` .
 - Routes are not mounted in `src/app.ts` .
 - Only `PreorderModel` exists in `src/modules/preorders/models/preorder.model.ts` (created for Phase 15 reporting aggregations), but its status enum (`['pending', 'confirmed', 'cancelled', 'fulfilled']`) deviates from Plan Section 31.1 (`['requested', 'admin_review', 'accepted', 'rejected', 'payment_pending', 'payment_verification', 'confirmed', 'available', 'fulfilled', 'cancelled']`).
- **File:** `Back-end/src/modules/preorders/`
- **Why it fails:** The pre-orders module was scaffolded during directory creation and model creation in Phase 15 for reporting, but customer request and administrative review endpoints were never implemented.
- **Recommended fix:** Implement `preorder.schema.ts` , `preorder.service.ts` , `preorder.controller.ts` , and `preorder.routes.ts` , mount routes in `src/app.ts` , update `PreorderModel` status enum to align with Plan Section 31.1, and add unit/API tests.

Failure 4: Unimplemented WhatsApp Integration Helper & Redaction Tests

- **Requirement:** Plan Section 38.1 and Section 63 (WA-001-003) mandate providing a safe WhatsApp link generation utility `buildWhatsAppUrl(phone, context)` with URL encoding and zero PII/secret leakage.
- **Expected:** `src/integrations/whatsapp/` contains the `buildWhatsAppUrl` function and associated URL encoding/redaction tests.
- **Actual:** `src/integrations/whatsapp/` contains only `.gitkeep` . `buildWhatsAppUrl` does not exist.
- **File:** `Back-end/src/integrations/whatsapp/`
- **Why it fails:** WhatsApp integration was identified as a client-side action MVP in Phase 0/Phase 17, but the shared helper function was never committed to `src/integrations/whatsapp/` .
- **Recommended fix:** Implement `buildWhatsAppUrl` in `src/integrations/whatsapp/whatsapp.adapter.ts` adhering strictly to Plan Section 38.1, and add unit tests verifying URI encoding and absence of sensitive tokens.

12. NEEDS VERIFICATION

Requirement	Why Verification is Needed	Missing Evidence	Artifact / Test That Should Prove It
Admin COD Reminder Endpoint (<code>POST /admin/orders/:reference/cod-confirmation-reminder</code>)	Plan Section 48.5 lists an endpoint to trigger an outbox reminder for COD confirmation. In Express routes, <code>POST /api/v1/orders/:reference/confirm-cod</code> exists for customer confirmation, but the admin trigger endpoint is not registered.	No admin COD reminder route registered in <code>admin-order.routes.ts</code> .	<code>admin-order.routes.ts</code>
OpenAPI Route-Sync CI Check	Plan Section 49.2 requires CI to fail when a route is registered in Express but absent from OpenAPI.	No automated test in <code>tests/</code> currently asserts route synchronization against <code>src/config/openapi.ts</code> .	Dedicated test <code>tests/contract</code> or <code>tests/security</code>
Admin Settings Key Mutation (<code>PATCH /admin/settings/:key</code>)	Plan Section 48.8 mentions <code>PATCH /admin/settings/:key</code> for dynamic configuration updates. Settings are currently handled via domain models and environment variables rather than a dedicated generic settings key-value store.	No <code>/admin/settings</code> route exists in Express router stack.	<code>app.ts</code>

13. OPEN DECISIONS (Preserved System-Wide)

In accordance with Plan Section 65, the following 24 items represent open business policy decisions owned by project stakeholders and are **not classified as implementation bugs**:

ID	Open Decision Item	Impact	Owner
OD-01	Final website phone number	WhatsApp deep links and footer contact	Store Owner
OD-02	Exact email-verification gating scope	Determines which account actions are blocked	Owner + Product
OD-03	Shipping rate source/logic & courier API	Real-time shipping calculation	Owner + Operations
OD-04	When final shipping cost binds & reconfirmation policy	Prevents cost disputes on manual adjustments	Owner + Product
OD-05	Delivery time ranges	Avoids unsupported delivery timing promises	Operations
OD-06	Excluded areas / serviceability policy	Shipping acceptance boundaries	Operations
OD-07	Reservation expiry policy & duration	When unaccepted reserved stock is released	Owner + Operations
OD-08	Fairness rule for competing last units	Allocation sequence for concurrent orders	Owner
OD-09	Low-stock / reorder thresholds	Inventory reorder warning levels	Owner
OD-10	Pre-order price honoring on catalog price change	Price snapshot vs catalog price at fulfillment	Owner
OD-11	Allocation order across multiple pre-orders	Fairness sequence when new stock arrives	Owner
OD-12	Required dynamic fields per service type	Schema configuration for custom services	Owner + Service Staff
OD-13	Service turnaround commitments	Customer SLA messaging	Service Staff
OD-14	Cash-on-Delivery applicability for service quotes	Service payment options	Owner
OD-15	Category-specific return periods and exclusions	Return eligibility policy	Owner
OD-16	Refund processing SLA duration	Accounting turnaround expectation	Owner
OD-17	Coupon stacking, limits, expiry, and minimums	Discount governance rules	Owner
OD-18	Business KPI target values	Performance reporting benchmarks	Owner
OD-19	Operational notification/email recipient list	Admin alert distribution	Owner + Operations
OD-20	Final payment-method list (Vodafone Cash / Instapay / etc.)	Manual payment instructions	Owner

ID	Open Decision Item	Impact	Owner
OD-21	Partial order acceptance behavior	Policy for partially available multi-item orders	Owner
OD-22	Payment-proof retention duration & deletion policy	Privacy and Cloudinary storage management	Owner
OD-23	Backup RPO, RTO, and snapshot retention	Recovery configuration	Owner + Hosting
OD-24	Exact post-acceptance non-financial order edits	Address/phone change rules after review	Owner + Operations

14. FINAL ACTION LIST

Must Fix (Actual Implementation Gaps)

1. OpenAPI Specification:

- Complete documentation in `src/config/openapi.ts` for the 96 missing Express routes.
- Add automated test verifying that every route registered in Express exists in OpenAPI.

2. Postman Collection:

- Generate and commit `postman/al-azhari-library.postman_collection.json` containing complete endpoint requests and environment variables.

3. Pre-orders Feature Module:

- Implement `preorder.schema.ts`, `preorder.service.ts`, `preorder.controller.ts`, and `preorder.routes.ts`.
- Mount pre-order routes under `/api/v1/products/:slug/pre-orders`, `/api/v1/pre-orders`, and `/api/v1/admin/pre-orders`.
- Align `PreorderModel` status enum with Plan Section 31.1.
- Add automated unit and API integration tests for pre-order workflows.

4. WhatsApp Link Helper:

- Implement `buildWhatsAppUrl(phone, context)` in `src/integrations/whatsapp/` adhering to Plan Section 38.1.
- Add unit test verifying URL encoding and parameter security.

Must Verify (Verification Items)

1. Verify whether an Admin COD reminder endpoint (`POST /admin/orders/:reference/cod-confirmation-reminder`) is required for launch or whether standard notification polling suffices.
2. Verify whether a dynamic settings key-value endpoint (`PATCH /admin/settings/:key`) is required or if environment/database configuration fulfills current operations.

No Action Required (Demonstrably Compliant)

1. Authentication, Sessions & Argon2id Password Hashing (`AUTH-001-008`) — 100% PASS
2. Product Catalog, Categories & Arabic Search Normalization (`CAT-001-008`) — 100% PASS
3. Shopping Cart Persistence, Optimistic Versioning & Merging (`CART-001-005`) — 100% PASS
4. Checkout, Orders, Multi-document Transactions & Idempotency (`CHK-001-012`, `ORD-001-009`) — 100% PASS
5. Manual Payments, Proof Uploads & Cloudinary Privacy (`PAY-001-008`) — 100% PASS
6. Atomic Inventory Reservations & Stock Ledger (`INV-001-008`) — 100% PASS
7. Student Services & Quotation Lifecycle (`SRV-001-008`, `QUOTE-001-005`) — 100% PASS
8. Returns & Manual Refunds (`RET-001-004`, `REF-001-003`) — 100% PASS
9. In-App Notifications & Socket.IO Realtime (`NOT-001-004`, `RT-001-003`) — 100% PASS

10. Transactional Outbox Pattern & Background Workers (OUTBOX-001-005) — 100% PASS
11. Content / CMS Seasonal Modules (CMS-001-004) — 100% PASS
12. Immutable Audit Logging & Analytics Reports (AUD-001-004 , REP-001-002) — 100% PASS
13. Security Hardening (Helmet, CORS, NoSQL Sanitizer, 6 Rate Limiters) (SEC-001-006) — 100% PASS
14. Database Migrations (13 versioned, idempotent migrations) — 100% PASS
15. Deployment Health Probes & Graceful Shutdown — 100% PASS
16. All 24 Open Decisions (Preserved without hardcoded assumptions) — COMPLIANT

15. Exact Command Evidence

```
# 1. Type validation
$ npx tsc --noEmit
Exit code: 0 (PASS)

# 2. Linting
$ npm run lint
Exit code: 0 (PASS, 0 errors, 0 warnings)

# 3. Clean production build
$ npm run build
Exit code: 0 (PASS -> dist/ generated cleanly)

# 4. Full test suite execution
$ npm test
Test Suites: 137 passed, 137 total
Tests:       938 passed, 938 total
Snapshots:   0 total
Time:        63.703 s (PASS)

# 5. Dedicated deployment smoke tests
$ npm test -- tests/deployment
Test Suites: 6 passed, 6 total
Tests:       25 passed, 25 total (PASS)

# 6. Dedicated security test suites
$ npm test -- tests/security
Test Suites: 14 passed, 14 total
Tests:       55 passed, 55 total (PASS)

# 7. Dependency vulnerability audit
$ npm audit
3 vulnerabilities (2 moderate, 1 high) – isolated, reviewed, documented, and mitigated in Outbox worker isolation
```

This concludes the Final Implementation Plan Compliance Audit for the AL-AZHARI LIBRARY backend.